

YUNING CHEN

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EDUCATION

University of California, Merced	Aug. 2021 - present
<i>Ph.D. Student; Working as Graduate Research Assistant and Teaching Assistant</i>	<i>Merced, CA, USA</i>
Tsinghua University	Aug. 2017 - Jun. 2021
<i>B.Eng. in Software Engineering; Excellent Bachelor Graduate(10%);</i>	<i>Beijing, China</i>

PUBLICATION

Kang Yang*, Yuning Chen*, Xuanren Chen, and Wan Du. Link Quality Modeling for LoRa Networks in Orchards. (* denotes equal contribution) ACM/IEEE IPSN 2023

RESEARCHES AND PROJECTS

Link Quality Modeling for LoRa Networks in Smart Agriculture. Aug. 2022 - Dec.2022
Graduate Research Assistant, Advisor: Prof. Wan Du, University of California, Merced

- Assesed the propagation modeling problem of wireless LoRa signals in smart orchard applications and explained the root drawback of existing path loss models.
- Proposed an interpretable propagation model for LoRa networks in orchards which leverages the first Fresnel zone theory and the uniform tree layout of orchards to model the shadowing effect caused by tree canopies and the ground with minimal overhead. Conducted extensive in-field experiments.

Application-layer optimization for Video Streaming over Multipath Sep. 2021 - Jul.2022
Graduate Research Assistant, Advisor: Prof. Wan Du, University of California, Merced

- Deployed a prevalent multipath transmission protocol(MPTCP) on Linux PCs and AWS EC2 servers to simultaneously communicate through WiFi and cellular. Accessed the QoE of different combinations of transport-layer mechanisms and application-layer adaptive bitrate algorithms.
- Proposed a DNN-based method to extract various features from 2 layers to predict transmission time more accurately thus achieving better selections and higher QoE.

Low-Latency Real-Time Video Streaming with WebRTC Jun. 2020 - Jan. 2021
Undergraduate Research Assistant, Advisor: Prof. Mo Li, Nanyang Technological University

- Built a live video streaming platform based on peer-peer connection, aiming to simulate and evaluate the real performance of WebRTC streaming, including signaling design and implementation.
- Implemented the Selective Forwarding Unit, a media server based on Janus to simulate the SOTA real-time video conference app; provided support for both PC and Android.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, Javascript, LaTeX, SQL, Golang, Matlab
Technologies	Pytorch, Git, MySQL, Linux, WebRTC, FFMPEG, Wireshark, Scikit-learn, NS-3

HONORS AND AWARDS

Awards

- ACM SIGBED student travel award (for ACM/IEEE CPS-IoT Week), 2023
- ACM CoNEXT travel grant, 2021
- Excellent Bachelor Graduate, 2021

- First Prize of Tsinghua University Student Research Training (SRT) Program Excellent Project (topic: weakly supervised learning), 2019
- First Prize in *Xin-Dong Project* Winter Camp (design and implementation of vision-radar hybrid smart vehicles based on MCU), Tsinghua University, 2018
- Outstanding Winner in the 7th Creative Competition · Junction X Tsinghua Hackathon, 2018

Scholarships

- Scholarships Indicating Comprehensive Excellence, Tsinghua University, 2018, 2019, 2020
- Bobcat Fellowship, University of California Merced, 2022